A MINI-PROJECT REPORT ON

AGRIPREDICT AI

Smart Agriculture Analytics & Multi-Modal Crop Prediction Platform

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In Partial Fulfillment of the Requirements for the Course:

DATA VISUALIZATION USING PYTHON (DVP)

Submitted Under the Esteemed Guidance of:

PROF. ANITA THENGADE

Department of Computer Engineering / Artificial Intelligence & Data Science

Submitted By:

Sr. No.	Student Full Name	PRN / Roll Number
1	**Anurag Kakad**	**1262241078**
2	**Shashwat Bakare**	**1262241168**
3	**Bijin Binu**	**1262241084**

Department of Artificial Intelligence & Data Science
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EXECUTIVE SUMMARY / ABSTRACT

Agriculture is the primary livelihood for over 54% of India's population. Despite its foundational role, farmers constantly struggle with climate variability, soil nutrient depletion, unbalanced chemical fertilization, and volatile market prices. **AgriPredict AI** is an enterprise-grade agricultural analytics and predictive intelligence platform developed for the **Data Visualization Using Python (DVP)** course under the academic supervision of **Prof. Anita Thengade**. Developed by **Anurag Kakad**, **Shashwat Bakare**, and **Bijin Binu**, the platform synthesizes five real-world datasets: (1) 246,091 national crop production records (1997–2015), (2) 771,456 hourly weather observations across 8 cities (2009–2020), (3) 620 laboratory soil chemistry samples, (4) 99 chemical fertilizer recommendations, and (5) 122 APMC Mandi commodity market prices.

Using Python, Streamlit, Scikit-learn, and Plotly, AgriPredict AI delivers responsive interactive visualizations (including choropleths, dual-axis climatograms, 11-variable polar radar plots, and price spread distributions) alongside benchmarked machine learning models: **95.97% accuracy** on soil crop recommendation (Random Forest), **100% accuracy** on fertilizer formulation, and an **R^2 of 0.8756** (RMSE: 3.406 Tonnes/Ha) on crop yield regression. The platform also features an APMC Mandi policy audit module and automated ReportLab PDF synthesis.

CHAPTER 1: INTRODUCTION, MOTIVATION & PROBLEM STATEMENT

1.1 Context and Background

India's agricultural landscape is characterized by tremendous regional diversity and seasonal dynamism. Cultivation cycles are structured around three primary seasons: **Kharif** (monsoon-dependent crops like Rice, Cotton, and Maize), **Rabi** (winter crops like Wheat, Barley, and Mustard), and **Summer/Zaid** (irrigated short-duration crops like pulses and vegetables), along with perennial whole-year crops such as Coconut and Sugarcane. Over 86% of Indian cultivators are smallholder and marginal farmers with landholdings under 2 hectares, making them highly vulnerable to economic and climatic shocks.

1.2 Motivation

Modern agricultural decision-making requires multi-dimensional intelligence: matching local soil chemistry with optimal crop varieties, selecting balanced chemical fertilizer formulations, understanding meteorological exposure and monsoon timing, and aligning crop production volumes with Mandi market prices and Government Minimum Support Prices (MSP). Traditional approaches suffer from critical information gaps:

- **Soil-Crop Mismatch:** Crop choices often follow historic intuition rather than laboratory soil chemistry.
- **Over-Fertilization:** Subsidized synthetic Urea is heavily over-applied, degrading soil health and lowering long-term yield.
- **Meteorological Anomalies:** Heatwaves and delayed monsoon rainfalls severely threaten crop phenology.
- **Market Opacity:** Farmers lack transparent real-time data on price realization and spread margins across APMC mandis.

1.3 Problem Statement

The central problem addressed by this project is the lack of an integrated, interactive computational tool that ingests disparate agricultural datasets, cleans and standardizes them, visualizes patterns dynamically, and provides predictive machine learning guidance for Indian farmers, students, and policy researchers.

1.4 Project Objectives

Under the guidance of **Prof. Anita Thengade**, our team set the following primary objectives:

1. **Multi-Dataset Ingestion:** Ingest 5 authentic Indian agricultural datasets without synthetic fabrication.
2. **Data Quality Engineering:** Clean missing records using crop-specific medians and detect IQR outliers.
3. **Interactive Plotly Visualizations:** Develop an exploratory visual studio for production, weather, soil, and market data.
4. **Machine Learning Benchmarking:** Evaluate Random Forest, Decision Tree, Gradient Boosting, and Regressors.
5. **Policy Audit & Reporting:** Benchmark open-market prices against MSP and automate PDF report generation.

CHAPTER 2: LITERATURE SURVEY & COMPARATIVE ANALYSIS

2.1 Existing Agricultural Decision Systems

Several existing systems and tools attempt to assist agriculturalists in India. Government portals such as Agmarknet and the Farmer Portal provide extensive records of market arrivals and weather bulletins, but they do so via static tabular reports that lack exploratory visual capabilities or real-time ML prediction. Conversely, academic research papers frequently focus on isolated machine learning algorithms run via terminal scripts or Jupyter notebooks, rarely wrapping their models in an accessible, user-friendly interactive web studio.

2.2 Shortcomings of Traditional Tools

Evaluation Dimension	Existing Portals / Scripts	AgriPredict AI (Proposed)
Interactive Visuals	Static HTML tables or static Matplotlib PNGs	Dynamic, responsive Plotly charts with zoom and tooltips
Data Scope	Single domain (either weather OR prices)	Unified 5-dataset integration (Crops, Weather, Soil, Fertilizer, Mandi)
Missing Value Handling	Naive zero dropping or global mean imputation	Crop-stratified median imputation and IQR outlier audit
Real-Time Inference	Requires technical coding environment	Interactive browser forms with confidence probabilities
MSP Policy Audit	Prices listed in isolation	Diverging delta charts flagging sub-MSP market distress
Automated Reporting	Manual copy-pasting or static PDF downloads	Automated ReportLab PDF compilation and CSV data packages

2.3 The AgriPredict AI Advantage

By packaging the full workflow—from raw CSV ingestion and cleaning to interactive charts, ML benchmarking, and executive PDF generation—into a modular Streamlit application, AgriPredict AI establishes a complete, transparent, and reproducible framework for academic and practical analysis.

CHAPTER 3: SYSTEM ARCHITECTURE & TECHNOLOGY STACK

3.1 Technology Stack Selection

AgriPredict AI is engineered using modern, robust open-source Python libraries:

- **Python 3.10+ / 3.14:** Core programming language powering all statistical and data workflows.
- **Streamlit (v1.35+):** High-performance web application framework for reactive, session-aware UI components.
- **Plotly Express & Graph Objects:** Interactive charting engine delivering hardware-accelerated SVG/Canvas visualizations.
- **Scikit-learn & Joblib:** Model training, ColumnTransformer feature engineering, and model serialization.
- **Pandas & NumPy:** Vectorized DataFrame operations, schema cleaning, and linear algebra routines.
- **ReportLab:** Programmatic PDF document compilation engine for executive summary generation.
- **Pytest:** Automated testing framework ensuring continuous pipeline integrity.

3.2 High-Level Architecture

The platform follows a decoupled Model-View-Controller (MVC) pattern, separating raw data ingestion from analytical rendering and user interaction:

Directory / Module	Primary Functionality & Contents
app.py	Main landing page, high-level KPIs, data health cards, and platform overview.
config/settings.py	Global design tokens, color constants, dataset paths, and column schemas.
preprocessing/	Resilient CSV loaders, median imputation, IQR outlier detection, weather aggregation.
analysis/	Domain analytics for crop trends, weather climatology, soil chemistry, and APMC prices.
models/	Crop classifiers, fertilizer ColumnTransformer pipelines, yield regressor, K-Means.
utils/charts.py	Reusable Plotly charting wrappers adhering to the Forest Emerald theme.
pages/	11 sub-pages implementing specific analytics, visual chart builders, and ML labs.
reports/	Export destination for generated project reports, PDF summaries, and documentation.
tests/	Automated unit test suite verifying loading, imputation, joins, and model inference.

CHAPTER 4: MULTI-MODAL AGRICULTURAL DATASETS

4.1 Dataset Inventory Breakdown

AgriPredict AI operates on five real-world datasets without synthetic fabrication, managed under strict governance policies:

Dataset Name	Record Count	Features	Key Attributes & Domain Scope
1. Crop Production	246,091 records	7 cols	State, District, Year, Season, Crop, Area, Production (1997–2015)
2. Hourly Weather	771,456 records	9 cols	8 Cities (2009–2020); Temp, Humidity, Precip, Pressure, Wind
3. Soil Chemistry	620 samples	12 cols	N, P, K, pH, EC, S, Cu, Fe, Mn, Zn, B; 6 Target Crop Classes
4. Fertilizer Rec.	99 instances	9 cols	Soil Type, Crop, Moisture, Temp, N, P, K; 7 Formulation Classes
5. APMC Mandi Prices	122 commodities	8 cols	Commodity, Variety, Min, Max, Modal Price (■/Qtl), Official Govt MSP

4.2 Detailed Profiles of Primary Datasets

1. Historical Crop Production: Sourced from the Directorate of Economics and Statistics (DES), Ministry of Agriculture. Contains 246,091 records spanning 36 Indian states and union territories, 124 distinct crop varieties, and cropping seasons from 1997 to 2015.

2. Meteorological Observatories: 11 continuous years (2009–2020) of hourly weather observations across Bengaluru, Bombay (Mumbai), Delhi, Hyderabad, Jaipur, Kanpur, Nagpur, and Pune. Captures daily thermal cycles, monsoon precipitation, and extreme heatwaves.

3. Soil Nutrients & Chemistry: Certified agricultural chemistry laboratory tests measuring primary macronutrients (N, P, K), soil acidity/alkalinity (pH), salinity (Electrical Conductivity), and 6 micronutrients (S, Cu, Fe, Mn, Zn, B) across 6 target crop types.

4. Fertilizer Formulations: Agronomic prescriptions mapping soil conditions, moisture, and crop variety to 7 balanced chemical fertilizer recipes.

5. APMC Mandi Market Prices: Cross-sectional agricultural market records documenting modal transaction prices alongside official Government MSP.

CHAPTER 5: DATA CLEANING, PREPROCESSING & QUALITY ENGINEERING

5.1 Ingestion with Multi-Encoding Fallbacks

Different regional agricultural databases in India employ inconsistent text encodings. Our data loader (`preprocessing/data_loader.py`) implements an automated fallback mechanism: it attempts `UTF-8` first, followed by `Latin-1` and `CP1252`, ensuring robust data loading without operating system crashes.

5.2 Grouped Median Imputation by Crop Category

In `crop_production.csv`, 3,730 records (1.51%) contained missing values in the `Production` column. A naive global mean or zero imputation would severely distort production estimates because agricultural outputs differ by multiple orders of magnitude across crop varieties (e.g., Sugarcane produces ~70 Tonnes/Ha, while Cardamom produces ~0.1 Tonnes/Ha). We implemented a crop-stratified median imputation:

$$Production_i = Median(\{Production_j \mid Crop_j = Crop_i \wedge Production_j \neq NaN\})$$

If an entire crop category lacked valid records, the national overall median was applied as a fallback.

5.3 Dynamic Feature Derivation: Calculated Yield

To evaluate land productivity fairly across large commercial farms and smallholder plots, continuous agricultural yield is dynamically derived:

$$Yield\ (Tonnes / Hectare) = Production\ (Tonnes) / Cultivated\ Area\ (Hectares)$$

Records with non-positive cultivation areas (Area ≤ 0) were removed as erroneous logging entries.

5.4 Interquartile Range (IQR) Outlier Diagnostics

Agricultural harvest distributions contain extreme values during bumper crop years or in intensive agricultural regions. Blindly purging outliers would discard critical domain reality. Our pipeline isolates and counts statistical outliers using the Interquartile Range:

$$IQR = Q■ - Q■ \quad | \quad Lower\ Bound = Q■ - 1.5 \times IQR \quad | \quad Upper\ Bound = Q■ + 1.5 \times IQR$$

Outliers are isolated, counted, and reported in the Dataset Governance Studio (`2_Dataset_Manager.py`) for user audit rather than silent deletion.

CHAPTER 6: DATA VISUALIZATION STUDIO (DVP CORE MODULES)

6.1 Core DVP Visual Dashboards

AgriPredict AI implements 11 specialized dashboard pages in Streamlit using Plotly Express and Graph Objects styled in an ergonomic Forest Emerald Dark Theme:

Dashboard Page	Visual Techniques Implemented	Core Analytical Insight Delivered
1. Executive Overview	KPI cards, Donut charts, Bar rankings	Macro indicators: Coconut (95.2% share) dominates national volume.
2. Dataset Governance	Health metrics, Schema tables, Null bars	Inspects data dimensions, missing percentages, and memory footprints.
3. Exploratory Studio	Correlation Heatmaps, OLS Scatters, Box plots	Custom chart builder with multi-variable color encoding and trendlines.
4. Crop Production	Spatio-temporal bars, State-Crop heatmaps	Filters production across states, crops, seasons, and years (1997–2015).
5. Weather Analytics	Dual-axis Climatograms, Monsoon Box plots	Rainfall vs. temp across 8 cities; counts heatwave & heavy rain days.
6. Soil Chemistry	11-Dimensional Polar Radar charts	Superimposes user soil nutrient chemistry against optimal crop bounds.
7. Fertilizer Analytics	Stacked Cross-Tabs, NPK 3D Scatters	Reveals disproportionate national over-reliance on chemical Urea.
8. Market Intelligence	Price Spread Candlesticks, Diverging Bars	Audits Mandi modal prices against MSP; flags sub-MSP market distress.
9. ML Model Lab	2D PCA Scatter, Confusion Matrices, ROC	Benchmarks classifiers, yield regressors, and K-Means clusters.
10. Executive Reports	ReportLab PDF synthesis, CSV packages	Single-click generation of formal publication-grade analytical reports.

6.2 Key Visualization Highlights

• **Dual-Axis Weather Climatogram:** Concurrently plots monthly rainfall column bars on the primary Y-axis with a red temperature spline curve on the secondary Y-axis, clearly visualizing monsoon onset.
• **Multi-Variable Polar Radar Plot:** Renders 11 chemical dimensions normalized to a common scale, enabling instant visual detection of nitrogen or potassium deficiencies.
• **MSP Policy Audit Diverging Bar:** Highlights commodities trading below official support prices in crimson, creating an immediate, intuitive policy audit tool.

CHAPTER 7: MACHINE LEARNING MODELS & PRESCRIPTIVE SYSTEMS

7.1 Supervised and Unsupervised Modeling Architecture

AgriPredict AI benchmarks multiple supervised classification, continuous regression, and unsupervised clustering algorithms implemented via `scikit-learn`:

Task / Target	Algorithm	Performance Metric 1	Performance Metric 2
Crop Recommendation (11 Soil Nutrients)	**Random Forest** Decision Tree Gradient Boosting	**95.97% Accuracy** 88.71% Accuracy 93.55% Accuracy	**0.9598 F1-Score** 0.8864 F1-Score 0.9351 F1-Score
Fertilizer Formulation (7 Chemical Classes)	**Random Forest Pipeline** Decision Tree Gradient Boosting	**100.00% Accuracy** 95.00% Accuracy 98.00% Accuracy	**1.0000 F1-Score** 0.9502 F1-Score 0.9801 F1-Score
Crop Yield Prediction (Tonnes / Hectare)	**Random Forest Regressor** Decision Tree Regressor Linear Regression	**0.8756 R^2 Score** 0.7912 R^2 Score 0.1843 R^2 Score	**3.406 Tonnes/Ha RMSE** 4.412 Tonnes/Ha RMSE 8.721 Tonnes/Ha RMSE
Soil-Climate Clusters (Unsupervised)	**K-Means (k = 4)** Principal Components	Silhouette Score: **0.462** Elbow Inflexion at k=4	PC1 Explained Var: 42.1% PC2 Explained Var: 24.8%

7.2 Real-Time Prescriptive Decision Support

• **Crop Suitability Inference:** Users enter their soil test parameters (N, P, K, pH, etc.) to receive top recommended crops along with probability distributions.
• **Fertilizer Grade Selector:** Users select their soil type, target crop, moisture level, and temperature to receive recommended fertilizer formulations.
• **Yield Estimation Tool:** Given a state, crop, season, and cultivation acreage, the regressor estimates expected harvest yield in Tonnes/Ha.

CHAPTER 8: KEY EMPIRICAL FINDINGS & AGRONOMIC INSIGHTS

8.1 Production Biomass Skew

Across 246,091 historical records, Indian agricultural output is overwhelmingly dominated by three crops: **Coconut**, **Sugarcane**, and **Rice**, which together account for over **96% of cumulative harvest weight**. Coconut alone accounts for over 129 billion units, concentrated heavily in Kerala, Tamil Nadu, and Andhra Pradesh. Conversely, high-value pulses and oilseeds account for less than 4% of total biomass despite occupying substantial land area.

8.2 The Urea Over-Reliance Dilemma

Cross-tabulation analysis in `8_Fertilizer_Analytics.py` reveals that **Urea** (46-0-0) represents over **35% of all fertilizer recommendations**, frequently prescribed across disparate soil types (Sandy, Loamy, Clayey). Balanced NPK formulations (e.g., 14-35-14, 20-20, 28-28) are critically underutilized, corroborating widespread agronomical warnings regarding declining soil nitrogen-use efficiency and soil acidification.

8.3 Regional Climatological Contrasts

Analysis of 11 years of hourly meteorological data across 8 regional hubs reveals stark climatic divergence:

- **Bombay (Mumbai):** Records the highest annual precipitation (1,223.7 mm), concentrated sharply in the June–September monsoon.
- **Kanpur & Jaipur:** Exhibit severe thermal stress, recording over 45 days/year exceeding 40°C heatwave thresholds.
- **Bengaluru:** Maintains the most stable climate with a 24.0°C annual mean temperature and moderate rainfall (624.6 mm).

8.4 Mandi Market Volatility and Distress Sales

APMC market price analysis shows that perishable cash crops and spices (e.g., Cardamom at ■97,000/Quintal) exhibit wide price spreads exceeding 40% between daily minimum and maximum quotes within the same market, reflecting heavy trader margins. Furthermore, comparing modal prices with Government Minimum Support Prices identifies sub-MSP market transactions for specific pulse varieties, demonstrating real-world farmer price distress.

CHAPTER 9: SYSTEM TESTING & AUTOMATED PYTEST VERIFICATION

To ensure continuous software reliability, an automated unit test suite was implemented in the `tests/` directory:

- **Data Loading Tests (`test_data_loading.py`):** Verifies file existence, schema integrity, and fallback encoding handling.
- **Data Cleaning Tests (`test_preprocessing.py`):** Verifies whitespace trimming, negative area removal, crop-specific median imputation, and calculated yield accuracy.
- **Model Inference Tests:** Verifies that serialized joblib models load reliably and return valid predictions and confidence probabilities.

All 12 unit tests execute and pass in under 1.5 seconds (`12 passed in 1.48s`), verifying that all core pipelines function correctly.

CHAPTER 10: LIMITATIONS, THREATS TO VALIDITY & DISCLAIMERS

10.1 Cross-Sectional Market Data Limitation

The APMC Mandi dataset represents a cross-sectional snapshot across 122 commodities rather than a multi-year daily time series. Longitudinal price forecasting (e.g., ARIMA or Prophet) was intentionally omitted to prevent misleading price projections.

10.2 Spatial Climatology Approximations

Meteorological data derives from 8 major urban stations. When joining weather records with district crop figures, urban stations serve as regional climate indicators for neighboring agricultural districts. Joins were strictly restricted to overlapping calendar years (2009–2015).

10.3 Agronomic Decision Support Disclaimer

AgriPredict AI is an academic decision-support framework developed for course evaluation. It is not intended to replace certified on-field soil testing, pedological laboratory analysis, or localized agricultural advisories from Krishi Vigyan Kendras (KVK).

CHAPTER 11: FUTURE ENHANCEMENTS & ROADMAP

- **Satellite Remote Sensing:** Ingest Sentinel-2 and Landsat multi-spectral imagery to compute real-time NDVI vegetation index maps.
- **Live Mandi REST APIs:** Connect the market intelligence module to Government of India `data.gov.in` Agmarknet live feeds.
- **IoT Microcontroller Telemetry:** Stream live soil moisture and temperature data from Arduino and ESP32 field telemetry nodes.

CHAPTER 12: CONCLUSION

The **AgriPredict AI** project successfully demonstrates how modern Python visualization and machine learning can transform complex, fragmented agricultural datasets into clear, actionable intelligence. Developed for the **Data Visualization Using Python (DVP)** curriculum under the esteemed guidance of **Prof. Anita Thengade** by **Anurag Kakad (1262241078)**, **Shashwat Bakare (1262241168)**, and **Bijin Binu (1262241084)**, the platform provides a comprehensive analytical foundation for precision agriculture in India.

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